

Fairness Has Memory

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Abstract

When a scarce, indivisible good is allocated by lottery among people with equal claims, the standard theory of fairness says the lottery is the fairest thing available. When the same kind of good is allocated the same way year after year, to largely the same people, the standard theory has nothing further to say: each year's lottery is fair, so the sequence is fair. I argue that this is a mistake. A sequence of individually fair lotteries can be an unfair allocation, and fairness requires that a claimant's chances in later rounds depend on the history of her unsatisfied claims. The argument uses only premises the standard theory already accepts: that a divisible good must be divided in proportion to claims, and that a lottery is a second-best surrogate for division. Over a sequence, the good becomes divisible. I then show what memory should track and how heavily it may press. It tracks a claimant's shortfall from her fair share, in units of the good where later units are as good as earlier ones and in waiting where they are not; and it may press that shortfall only as hard as is consistent with every claimant's still receiving, in each period, the surrogate she is owed. Rules that forget, like New Mexico's hunting draw and the United States visa lotteries, and rules that remember lexically, like preference-point queues, are both unfair, in opposite directions. Finally I say how far back, and how far across, fair memory reaches.

I Two lotteries

Every year Nevada and New Mexico each have more people who want to hunt bighorn sheep than they have sheep to spare, so each holds a lottery. The two lotteries differ in one respect that will occupy this paper. Nevada remembers. An applicant who is unsuccessful earns a bonus point; her points are squared to fix how many random numbers are drawn for her the next year, and the lowest number she draws is her entry, so ten years of losing gives her a hundred and one draws against a newcomer's one.¹ New Mexico forgets. Its Department of Game and Fish states the policy in one sentence: "New Mexico does NOT grant preference to applicants who were unsuccessful in previous drawings."² A hunter passed over for ten years enters the eleventh draw with exactly the chances of someone entering for the first time.

Which of these is fair? Most people, asked, side with Nevada. When Huesch and Brady (2010) asked

¹Nevada Department of Wildlife, "Bonus Point Program," <https://www.ndow.org/blog/bonus-point-program/>. Points revert to zero when the applicant draws a tag, or after two consecutive years of not applying.

²New Mexico Department of Game and Fish, "How the New Mexico Draw Works," <https://wildlife.dgf.nm.gov/hunting/applications-and-draw-information/how-new-mexico-draw-works/>.

subjects to design a pair of back-to-back raffles among the same ten people, 78 per cent said the winner of the first should be excluded from the second, and 75 per cent said so even when asked as disinterested spectators. Many institutions side with Nevada too, though they disagree about how to remember. Arizona gives an unsuccessful hunter one extra random number per year rather than squaring them. In Colorado's and Wyoming's main draws, licences go to whoever has accumulated the most preference points, and chance only breaks ties. The Grand Canyon's river-permit lottery gives an applicant one ticket for each year since she last held a permit, to a maximum of five. Kidney allocation in the United States is a points system whose principal component is waiting time. Publicly funded IVF in England is capped at a fixed number of cycles per person.³ And then there are the systems that forget: New Mexico, the Wimbledon ballot, and, most consequentially, the lotteries by which the United States allocates the roughly 85,000 H-1B visas Congress permits each year and the 55,000 Diversity Visas. A skilled worker whose registration was not selected in three successive years enters the fourth draw with the chances of a newcomer. That remains true under the rule that took effect in February 2026, which weights H-1B registrations by the wage level of the job offered, entering the highest-paid four times and the lowest once, but gives no weight to how many times a person has already been passed over.⁴

The theory of fair allocation has not connected this disagreement to the theory of claims, and where it touches it, it sides with New Mexico. The best-developed account, due to John Broome (1984a, 1991a, 1991b), holds that fairness requires people's claims to a good to be satisfied in proportion to their strength, and that when the good is indivisible and claims are equal, an equal lottery is the fairest available procedure. The theory evaluates a lottery on its own. On its own, each year's New Mexico draw is impeccable, and if the sequence is nothing but its members, the sequence is impeccable too. Broome's critics, who deny that chances are a kind of satisfaction and locate the fairness of lotteries in what they express or exclude (Wasserman 1996; Stone 2007, 2011; Spiekermann 2022), reach the same verdict by a different route: an impartial procedure that treats everyone identically is fair, whatever last year's procedure did. The wider literature on lotteries (Sher 1980; Kornhauser and Sager 1988; Elster 1989; Duxbury 1999; Saunders 2008; Stone 2008) takes the single lottery as its unit; Goodwin's (2005) "total social lottery," which periodically redistributes society's goods, is the forgetting rule adopted as a constitution. The recent literature on queues (John and Millum 2020; Hersch and Rowe 2024) asks whether waiting should count, which is a question about history, but does not connect the answer to claims. The nearest things to the thesis I will defend are two passages by Hersch and Rowe. In one, a parent hands out a chocolate bar every ten minutes, and the child who got the first insists that

³Colorado Parks and Wildlife, "Primary Draw," <https://cpw.state.co.us/hunting/big-game/primary-draw>; Wyoming Game and Fish Department, "Understanding the Draw" (2024); Arizona Game and Fish Department, "Bonus Point Process"; Grand Canyon National Park, "Noncommercial River Trip FAQ," <https://grcariverpermits.nps.gov/faq.cfm>; National Academies of Sciences, Engineering, and Medicine, *Realizing the Promise of Equity in the Organ Transplantation System* (2022), ch. 5; NICE guideline NG257, "Fertility problems: assessment and treatment" (2026), recommendations 1.39.6–1.39.9. The NBA draft lottery, weighted by the previous season's record, and the Amsterdam housing register, which awards a point per year registered and a further point per month of unsuccessful applications, also remember; the Wimbledon public ballot and most American school-admission lotteries do not.

⁴Department of Homeland Security, "Weighted Selection Process for Registrants and Petitioners Seeking To File Cap-Subject H-1B Petitions," 90 *Federal Register* 60864 (29 December 2025), effective 27 February 2026.

“chocolate bar at t_1 is not the same good as chocolate bar at t_2 , and a lottery must be had for this new and distinct good”; they reply that if we index goods to times “we lose something that matters for fairness,” without saying what (Hersch and Rowe 2024, sec. 5). In the other, a survey, they observe that recurring chores are “not independent of each other,” that a recurring chore “which extends temporally” is not indivisible even when each instance is, and that “who completed the chore last time matters to who has a claim to avoid it this time” (Hersch and Rowe 2025, 3). Both observations are right. This paper says what is lost, why, how far it reaches, and what follows for the institutions just described.

The thesis is that fairness has memory. More precisely, I will argue for three claims.

Non-separability. The fairness of a sequence of allocations is not a function of the fairness of its members as the standard theory assesses them. A sequence of lotteries each of which is fair, on that theory, can be and typically is an unfair allocation.

Memory. When a good is allocated repeatedly among overlapping claimants, fairness requires that each claimant’s chances in later rounds depend on her history of unsatisfied claims.

Currency and weight. What memory should track is a claimant’s shortfall from her fair share, in the currency in which her claim is divisible: units of the good, where a later unit is as good to her as an earlier one; waiting, where it is not. And it may press that shortfall only as hard as is consistent with every claimant’s still receiving, in each period, the surrogate she is owed. A rule that forgets fails the first requirement. A rule that lets history decide, so that a newcomer’s present claim counts for nothing, fails the second. Both are in use.

The argument for the first two claims (Section 3) needs nothing beyond premises that Broome and most of his critics already accept: that when a good can be divided among equal claimants, fairness requires that it be divided, and that a lottery for an indivisible good is an admittedly imperfect surrogate for division. Over a sequence, the good becomes divisible. The argument seems to conflict with the standard theory’s doctrine that losing a fair lottery extinguishes one’s claim; Section 4 shows that the conflict is only apparent. Section 5 turns to the third claim and to design. Section 6 asks how far back, and how far across, memory reaches: does a housing loss count in a visa lottery, a parent’s loss for a child, bad luck at the hands of no one at all? The answers are derived rather than stipulated, and they keep the thesis clear of luck egalitarianism and of theories of rectification. Section 7 answers the remaining objections.

2 Claims, chances, and division

Broome’s theory begins with claims. A claim, in his sense, is a duty owed to the candidate herself that she should have the good, as distinct from other reasons for giving it to her, such as that doing so would be useful (Broome 1991a, 90–92). Need is the paradigm source of claims; desert and prior agreement are others. Fairness is a specific requirement on how claims are treated: “equal claims require equal satisfaction,” “stronger claims require more satisfaction than weaker ones,” and “weaker

claims require some satisfaction. Weaker claims must not simply be overridden by stronger ones” (Broome 1991a, 95). Fairness is thus comparative, “concerned only with relative satisfaction, not absolute satisfaction” (Broome 1991a, 95); it can be perfectly met by denying the good to everyone, and the requirement that claims be satisfied as much as possible competes with it (Broome 1991a, 97).

Two features of the theory will bear the weight of my argument. The first is what I will call the Division Principle. Where a good is divisible, proportional satisfaction is straightforwardly achievable, and the theory says it is required. If two people have equal claims to a quantity of food, fairness says the food “should be divided,” even when giving it all to one of them would do more good (Broome 1991a, 95). Broome states the principle at its most general in a passage that already contains, in miniature, the idea that a good can be divided across time:

I have been implicitly assuming that the good to be distributed is indivisible. Sometimes we are faced with an indivisibility in the nature of things; the good, for instance, of staying in a lifeboat cannot be divided. But sometimes the problem of selection arises even with a good that could be divided. The good, for instance, of avoiding military service could be divided by conscripting everyone for a short time rather than a few people for a long time. If people’s claims to a good are equal, then fairness requires that the good should be divided between them if it can be. That is the only way to treat them with perfect equality. (Broome 1984a, 48)

The second feature is the account of lotteries as a second best. When the good really is indivisible, and there is not enough of it, “some unfairness is inevitable,” because “some candidates will get the good and others will not” (Broome 1991a, 97–98). What a lottery achieves is “a sort of partial equality in satisfaction”: “each person can be given a sort of surrogate satisfaction. By holding a lottery, each can be given an equal chance of getting the good. This is not a perfect fairness, but it meets the requirement of fairness to some extent” (Broome 1991a, 98). A chance counts as surrogate satisfaction because “if you have a chance of getting the good you may actually get it,” unlike merely giving a claim its due weight in a deliberation that goes against it (Broome 1991a, 98). And the surrogate is explicitly a surrogate: the result of a lottery among unequal claimants “will generally be that the good goes to candidates who do not have the strongest claims. This is less fair than the result of giving it directly to those who do” (Broome 1991a, 98–99). In the 1984 paper the lottery achieves “a fairness that is less than perfect,” and is what one does when division, the perfectly fair option, “may be very damaging to the general good” (Broome 1984a, 48). Two things follow that I will use. Where division is available, it is required. Where it is not, the surrogate is not merely permitted but owed: a claimant who cannot be given her portion must be given her chance.

The theory has been much criticized (Hooker 2005; Tomlin 2012; Lazenby 2014; Kirkpatrick and Eastwood 2015; Piller 2017), and weighted lotteries have been refined and contested chiefly in the debate about whether the numbers count (Kamm 1993; Broome 1998; Timmermann 2004; Henning 2015; Saunders 2010; Vong 2020; Tank et al. 2024). But the Division Principle and the second-best status of lotteries are common ground. Wasserman (1996) doubts that chances are a kind of satisfaction, since “hungry claimants” cannot “actually eat chances” (p. 47), and holds that among equally enti-

tled claimants “we have no reason save economy to equalize their expected rather than their actual outcomes” (p. 46); he also counts “time sharing” as literal division of a good (p. 30 n. 1). Stone (2007, 284–85; 2011) holds that a lottery is just because impartiality demands that some be favoured over others “only for legitimate reasons,” and a lottery keeps illegitimate ones out of a decision no legitimate reason can settle; but dividing a divisible good is not such a decision. Lazenby (2014, 341) holds that whatever fairness a lottery contributes is lexically posterior to fairness in outcome. My argument needs only what these positions share.

One more preliminary. The theory speaks of claims, and it may be asked what duty Nevada owes anyone to let her shoot a sheep. The answer is that the claims in the cases that concern me are claims from the allocator’s own undertaking. When a public body announces that it will allocate a public resource among eligible applicants by a fair procedure, it undertakes to each applicant that her application will be treated as the procedure promises; each eligible applicant thereby has a claim, owed to her, that is equal to every other’s unless the rules distinguish them. Agreement is one of Broome’s own sources of claims, and this is agreement of a public kind. It extends to a private allocator that advertises a ballot, and it is what makes the visa registrant’s position a matter of fairness rather than of policy alone. What it does not do is turn every reason for allocating a visa into a claim. The wage level of the job an H-1B beneficiary has been offered is a reason of the other kind, a reason of usefulness, and the 2026 reform that weights the lottery by it has made the lottery more responsive to usefulness, not to claims. I return to this in Section 5.4.

3 The divisibility argument

3.1 The setting

Consider a good that comes in indivisible units, one unit per period, allocated by a single agency among a set of claimants whose claims are equal, and where receiving a unit in one period does not remove a claimant from the pool: a bighorn tag a year, a river launch a season, a night off the rota a week. To begin with I assume the claimants are the same from period to period; I relax this in Section 5. Let there be N claimants and T periods. Within any one period the standard theory’s verdict is clear and, I think, correct: one unit, N equal claims, no way to divide the unit, so hold an equal lottery. The question is what the theory says about the T periods taken together.

3.2 The argument

The units are indivisible. The sequence of units is not. Over T periods, it is possible to give each claimant T/N units, or as near to that as the integers allow: if T is a multiple of N , exactly T/N each; otherwise some get one more than others, but nobody’s total differs from anybody else’s by more than one. Call the procedure that achieves this *rotation*: draw a random order once, and then cycle through it.⁵ Rotation divides the sequence among the claimants in proportion to their equal

⁵When T is not a multiple of N , the random initial order settles fairly who gets the extra unit. Rotation is not the only procedure with this property; excluding past winners until everyone has won and then re-drawing has the same

claims. It is, in the sequence, what dividing the food is in the single case.

The Division Principle says that “if people’s claims to a good are equal, then fairness requires that the good should be divided between them if it can be.” The sequence can be divided. So fairness requires that it be divided. So fairness requires rotation, or something that comes as close to it as circumstances allow.

Now compare the procedure New Mexico follows: a fresh, independent, equal lottery in every period. Call it the *forgetting* rule. In any one period it does what the theory asks. Over T periods it does not divide the sequence; it raffles it. With ten claimants and ten periods, the forgetting rule leaves, on average, three and a half of the ten with nothing at all, while two or three of the others receive two or more units. With a hundred claimants and a hundred periods, 37 of the hundred will never have won and 26 will have won twice or more.⁶ Rotation leaves nobody with nothing. The forgetting rule does over the sequence exactly what the theory forbids in the single case: it takes a divisible good and gives it out by lot.

Here is the point put as a reductio. Suppose someone proposed to allocate a cake among ten people with equal claims as follows: cut it into ten slices, and raffle each slice separately. Each raffle is, taken by itself, exactly what the theory recommends for an indivisible good. Yet nobody thinks this is a fair way to allocate a cake. The theory itself says the cake should be divided, and dividing it means giving everyone a slice, not giving everyone a chance at each slice. The only difference between the cake and the tags is that the slices are allocated simultaneously and the tags a year apart. If simultaneity is not what makes the cake divisible, and it is not, since a cake sliced today and handed out over ten days is still a divisible cake, then the tags are divisible too.

Two consequences follow. The first is the failure of separability. The standard theory assesses a lottery by the claims present when it is held, and it takes those claims to be fixed by need, desert, agreement and the like, not by what earlier lotteries did. Say that an allocation rule is *locally fair* at a period if the procedure it uses at that period, assessed in that way, is fair. The forgetting rule is locally fair at every period: at each draw, everyone present has an equal claim and an equal chance. Rotation is not locally fair at any period after the first, since given the history one claimant is certain to receive the unit and the others certain not to. Yet rotation is the fair allocation of the sequence and the forgetting rule is not. So local fairness at every period is neither necessary nor sufficient for the fairness of the sequence: fairness, assessed as the standard theory assesses it, is not separable across occasions. There is another way to put the same result, and from Section 4 on I will use it. If one lets claims carry history, so that a claimant who has received less than her share has, on that account, a stronger claim to the next unit, then rotation *is* locally fair at every period and the forgetting rule is not, and fairness is separable after all. Non-separability of fairness with respect to history-blind claims and memory in claims are one thesis in two vocabularies. What is not available is the standard theory’s combination: history-blind

distribution of totals. Nothing below turns on the difference.

⁶The expected number of claimants with nothing is $N(1 - 1/N)^T$; for $N = T = 100$ this is $100 \times 0.99^{100} \approx 36.6$. The other figures are computed in the same way, or by simulation as described in the Appendix.

claims and a separable notion of fairness.⁷

The second consequence is memory. Rotation is a rule under which who receives the unit in period t depends on who received it in periods before t . Any rule that divides the sequence must have this feature, since dividing the sequence means giving units to those who have received fewer, and that requires knowing who has received fewer. So if fairness requires division of the sequence, fairness requires memory.

3.3 What the argument assumes

The argument does not assume that chances are a kind of satisfaction. It assumes only that a lottery is a second best relative to division, which is common ground.

It does assume something that needs stating, because it is where a careful opponent will press. The Division Principle applies to a good that “could be divided.” In Broome’s examples the claimants end up with portions of one thing, and the tags a hunter draws in different years are not portions of one tag. What licenses treating them as portions of one good is a premise about the claim. Call it *Fungibility*: units allocated in different periods are portions of one divisible good, for a claimant, to the extent that her claim is to a unit of that kind and a later unit would satisfy it as an earlier one would. The hunter’s claim is to a sheep tag, not to the 2027 tag; the conscript’s, in Broome’s example, is to avoid service, not to avoid it this month. Where Fungibility holds, the sequence is divisible in the currency of the claims, not merely in the allocator’s bookkeeping.

Fungibility is a matter of degree, because delay is nearly always some cost, and it fails outright in two ways. It fails when the claim is to a particular unit: the one transplant organ that matches, the seat on the one lifeboat. Then there is no sequence and the standard theory applies as it stands. And it fails, more interestingly, when the claim is to a unit of the kind but a later unit is much worse for the claimant: the housing applicant who will be evicted before next year’s allocation, the visa applicant whose sponsor will not wait. Here the claim is largely a claim to timeliness. But timeliness is divisible: delay can be shared out among claimants as units cannot. In this case the argument does not lapse; it changes currency. Fairness requires that the *waiting* be divided in proportion to claims, and the memory this requires tracks time waited (Section 5.4). In the general case, where a later unit is worth something but less, the currency is units discounted by delay, and the pure cases are its endpoints.

The argument does not assume equal claims either. With unequal claims the sequence should be divided in proportion to them, and doing so with indivisible units is the apportionment problem; the divisor methods that solve it for parliaments have a sequential form, which is a rule with memory.⁸

⁷Broome’s theory of goodness is organized around separability principles for betterness across people, times, and states (Broome 1991b). Non-separability of fairness across occasions is a different kind of non-additivity, of a comparative pattern rather than a value, and I do not claim it bears on those principles. It does bear on how fairness is weighed against goodness: if fairness attaches to sequences, the weighing must be done at the level of the sequence.

⁸Wintein and Heilmann (2018) show that the divisor methods of apportionment theory divide several indivisible units proportionally without lotteries; the classic treatment is Balinski and Young (2001). In sequential form a divisor method gives each unit to the claimant who maximizes the ratio of her claim to the number of units she has already received plus a constant; with equal claims every divisor method reduces to rotation. The same point has lately been made in computer

4 The extinction objection

The argument so far might seem to run into a doctrine that Broome states in the same year as the Division Principle and that his followers have made central. When a fair lottery has been held and Michael has won the dialysis machine, it would be absurd, Broome says, to hold another lottery on the ground that lotteries are fair: “once Michael has won the first lottery he has a perfect claim to the treatment, and Maggie has none. Consequently it is not at all fair to hold a second lottery” (Broome 1984b, 628–29). Call this the *extinction thesis*: losing a fair lottery extinguishes the loser’s claim. Vong (2015), who notes that “there has been little discussion of the temporal structure of how lotteries satisfy beneficiaries’ claims” (p. 470), rejects the thesis in one form and keeps it in another. He distinguishes a *procedural* claim, “satisfied by a lottery that gives claimants appropriate, fair chances of benefiting,” from a *benefit* claim that “cannot be lost due to the results of a procedurally fair lottery” (pp. 479–480). The benefit claim survives losing; the procedural claim, once satisfied, is satisfied. So when new units become available, the losers are to be treated “as a new case of scarce benefit distribution” and a fresh fair lottery run among them (p. 483). He considers, and rejects, views on which losing a lottery weakens a claimant’s claim and on which winning strengthens it (pp. 477–479). Feldblyum Le Blevennec (2023) proposes a “weighted diachronic lottery” for cases in which new claimants arrive after a lottery has been held but before the good is handed over; but in her scheme, as in a proposal of Broome’s that Vong reports (p. 481), the losers of the first lottery are *excluded* from the second, and it is the winner whose chances are protected.

The objection comes in two forms. On Broome’s, the persistent loser in New Mexico has no more claim than anyone else: each year her claim was fairly entered, surrogately satisfied by an equal chance, and extinguished. On Vong’s, her benefit claim survives, but so does everyone else’s, equally, and what she is owed each year is a fresh fair lottery among all who remain, which is the forgetting rule. On neither is there anything for memory to track, and the existing literature on lotteries with memory, such as it is, has memory running the other way: it is winners whose history counts.

The reply is to distinguish three structures that the objection runs together.

The first is a *re-run*: the same claimants, the same unit, a second lottery held because the first went the wrong way for someone. Broome is right that this is absurd, and the reason is instructive: there is nothing left to divide. The loser’s claim to *that* unit was given its surrogate and the unit is now the winner’s; another lottery would take it back from someone with a perfect claim to it. The extinction thesis is correct about the re-run, and I accept it.

The second is the *interrupted allocation* that Vong and Feldblyum Le Blevennec discuss: one unit, a lottery held, and then new claimants who arrive before the unit is handed over. Here too there is one unit, and memory of *winning* is what the case calls for, since the problem is to protect the standing of a winner whose lottery was fair without treating late arrivals as though they had no claim. I have

science, where “repeated” and “temporal” fair division begins from the observation that an allocation of indivisible items that cannot be made envy-free in one round can be made envy-free over enough rounds (Igarashi et al. 2024; Aleksandrov and Walsh 2020).

no quarrel with this literature.⁹

The third structure is *recurrence*: a new unit each period, allocated among claimants who overlap from period to period. In this structure the extinction thesis is true and irrelevant. Maggie's claim to the 2026 tag was extinguished when she lost the 2026 lottery; it is someone else's. But there is a 2027 tag, and the Division Principle says that the 2026 and 2027 tags together are a divisible good, so that the 2027 tag goes, or is more likely to go, to someone who did not get the 2026 one. Maggie's claim to the 2027 tag is not the ghost of her claim to the 2026 tag. It is a claim to a unit that exists, and it is stronger than the claim of someone who received the 2026 unit for the same reason that, when a cake is being divided, the claim of someone who has had no slice yet is stronger than the claim of someone who has had one. Where the currency is waiting, the same holds with waiting in place of slices: her claim to be served now is stronger than a newcomer's because she has borne more of the delay being divided.

So the extinction thesis and the memory thesis are about different goods. What Broome's form of the objection needs, in order to bite, is a stronger thesis: that losing a fair lottery for one unit extinguishes not only the claim to that unit but any bearing the loss might have on the allocation of later units. And that stronger thesis is what the cake refutes. Someone who raffled each slice separately could say, of each raffle, that the losers' claims to that slice had been fairly dealt with and extinguished; it would still be true that the cake had not been divided. Vong's form fails for a related reason. A procedural claim is a claim to a fair procedure, and it is satisfied by a fresh equal lottery only if a fresh equal lottery is the fair procedure; over recurring units, the Division Principle says it is not. The option Vong does not consider, between the weakening and strengthening views he rejects, is that losing strengthens the claimant's claim to *later* units; and his objections to those views, which turn on the arrival of new claimants for a single unit, do not touch it. His own case against Broome is, in fact, a demand for memory of a modest kind: the time at which one fell overboard should not affect one's cumulative chance of the buoy, so later lotteries for one unit must be adjusted for earlier ones (pp. 474–476). The argument of this paper extends that demand from one unit to the sequence.

A residual worry is worth drawing out, because answering it is what makes the surrogate character of chances do real work. If the lottery gave Maggie surrogate satisfaction in 2026, then in 2026 her claim *was* satisfied, if only by surrogate; is extra weight in 2027 not satisfying it twice? This is where it matters that a surrogate is a surrogate. A chance is not a portion of the good; it is what one gives when a portion cannot be given. If it were a portion, so that an equal chance of a slice were a kind of equal slice, then the raffle of the cake would be a division of the cake, and the theory could not say, as it does, that the cake should be divided instead. The whole content of the Division Principle is that surrogate satisfaction is not to be preferred to real satisfaction when the real thing is available, and in the recurrence structure it is available over the sequence. So the surrogate Maggie received in 2026 is

⁹Vong (2012, 9–10) himself appeals to repetition, observing that when the same two groups meet in a conflict every month, always saving the greater number lets the smaller group lose every time, and taking this to favour a lottery over majoritarianism. That is an argument from repetition to lotteries; mine is one from repetition to memory in lotteries, and they are compatible.

no reason to withhold from her the 2027 unit when it can be given.¹⁰

5 What memory tracks

5.1 Two structures

Sort the institutions of Section 1 by structure before asking what fairness requires of them. In the *recurrence* structure, a claimant who receives a unit stays in the pool and may receive another: hunting tags, river permits, chores, ticket ballots, the annual lotteries some funders now use for research grants. In the *bottleneck* structure, a claimant who receives a unit leaves: a visa, a flat, a kidney, a school place. The structures are individuated by whether winners leave, the currency of memory by Fungibility, and the two are independent; but they run together in practice, since the claimant who leaves on being served was waiting for one unit and cared when it came. Mixed cases, a rota whose nights are not interchangeable, a grant that comes too late for the laboratory, have units discounted by delay as their currency; and a bottleneck whose claimants did not care when they were served would be one where no one is waiting for anything, the forgetting rule is fair, and the persistent loser has no complaint. I take the two pure cases in turn.

The designs in use track three things. *Points for losses* track how often a claimant has lost since she last won, and raise her chances linearly (Arizona), convexly (Nevada), or in reverse, by lowering the chances of recent winners (the Grand Canyon). *Queues* track how long a claimant has waited and serve whoever has waited longest, chance only breaking ties (Colorado, Wyoming, and in its principal term kidney allocation). *Caps* stop a claimant at a limit. And they differ in how hard they press what they track: a queue makes history decisive, a linear point system makes it a matter of degree.

5.2 Shortfall in units

In the recurrence structure the argument of Section 3 fixes what memory should track. Fairness requires that the sequence be divided in proportion to claims; the measure of a claimant's treatment is the difference between the share of the units she should have received by now and the number she has received. Call this her *shortfall*. With equal claims and one unit per period, a claimant present for r periods who has received w units has a shortfall of $r/N - w$. It can be negative; a claimant who has received more than her share has a surplus. It is measured in units, and it is signed.

Shortfall is neither the number of losses nor time waited. Points-for-losses systems reset to zero on a win, so that a claimant who has won once in twenty years is treated as a newcomer though her shortfall is large, and they count a loss in a year with two hundred applicants exactly as a loss in a year with two.

¹⁰The structure resembles what Thomson (1990) calls the residue of a justified infringement, with one difference that matters in Section 6: nothing was infringed. The lottery was the fairest thing available. What remains after it is not a wrong to be compensated but a division still to be completed. This also answers the objection that memory "double counts" chances. What is counted, on the view I am defending, is not chances but units received. Chances are the currency in which we pay when we cannot pay in units; they are not themselves entered in the ledger. Wasserman's (1996) remark that one cannot eat chances is, on this view, the beginning of an argument for memory rather than an objection to lotteries.

Time waited tracks shortfall only if the numbers of units and claimants are constant. What both get right, against the forgetting rule, is that a claimant's past bears on her present claim; what they get wrong is the quantity.

Consider, then, the family of rules that track shortfall directly and press it with a weight $\kappa \geq 0$: at each period, give each claimant a chance proportional to $1 + \kappa \cdot \text{shortfall}$, with any negative value set to zero. At $\kappa = 0$ this is the forgetting rule. As κ grows, a claimant's chance is increasingly dominated by her shortfall, and in the limit the rule gives the unit to whoever has the largest shortfall, which with equal claims is exclusion of past winners with random tie-breaking. At $\kappa = 1$, a claimant's expected shortfall shrinks by a factor of $1 - 1/N$ each period, so it is corrected over roughly the time the practice takes to hand out one round of shares.

The floor at zero is the mechanism behind the third claim. A claimant who has received more than her share cannot be given a negative chance; the most one can do is exclude her until the others catch up. This is the extinction thesis built into the rule: past winners keep what they won. But it introduces an asymmetry: shortfalls can be pressed upward without limit, surpluses only down to zero, and the more heavily κ presses, the more the clipped surpluses fail to offset the amplified shortfalls. In a fixed population this is harmless. Everyone starts level, so every claimant has the same chance of every unit whatever κ is, by symmetry, and as κ rises the dispersion of totals falls: with ten claimants and ten periods, the expected number left with nothing falls from 3.5 under the forgetting rule to 2.1 at $\kappa = 1$, 1.3 at $\kappa = 2$, and 0 at $\kappa = 10$. In a fixed population, then, the Division Principle says: press the shortfall as hard as you can, and rotate. The asymmetry bites only when claimants come and go.

5.3 Newcomers

The population is not fixed. Hunters take up the sport and abandon it; applicant pools turn over; households form and dissolve. And once claimants come and go, a real objection to memory appears that the fixed-population case hides. In the year in which she first applies, a newcomer has a claim exactly as strong as anyone else's. Under any rule with memory, she is given a smaller chance than an incumbent who has lost before. So the rule gives unequal chances to equal claims, which is what the standard theory says unfairness is. Memory, on this objection, corrects one unfairness by committing another, and the newcomer pays.

The objection has an answer, and the answer also fixes how heavily memory may press. Return to the two things I drew from the standard theory in Section 2: division is required where it is available, and where it is not, the surrogate is owed. Division of the sequence is available only to a claimant who stays long enough for it to be completed. A claimant who leaves after three periods cannot be given three-tenths of a tag; for her, the sequence is as indivisible as the unit. If the allocator knew who would leave, it would owe such claimants the surrogate outright: a chance, in each period, proportional to their claim. It does not know. So it owes the surrogate to everyone, in the only form in which it can be given without knowing tenures in advance: that no claimant's expected share, over whatever tenure she turns out to have, fall short of proportionality to her claims. Call this the *surrogate constraint*. It is what the second-best status of lotteries requires when the allocator cannot tell who will stay

long enough to be treated by division, and it is prior to the Division Principle’s demand for the same reason the surrogate is owed at all: fairness may reduce the unfairness of the sequence’s outcome only by means that do not deny anyone the chance her claim warrants. What fairness requires is the rule that presses shortfall as hard as the surrogate constraint allows.

Which rules those are is a matter of calculation. Suppose claimants leave with a fixed probability each period, independently of their history, so that tenure is random and unknown to claimant and allocator alike, with a mean of ten periods, and the pool is kept at ten by replacing those who leave. The question is what a claimant’s expected total is, as a function of how long her tenure turns out to be. To ask this is to take the claimant over her whole tenure, rather than at a period, as the unit fairness compares: the analogue of the “complete lives” view of equality that McKerlie (1989) distinguishes from views comparing simultaneous segments of lives (see also Temkin 1993, ch. 8), forced here because claims belong to persons, who persist through the periods, not to persons-at-periods.

Table 1 reports the results. Under the forgetting rule, expected share is proportional to tenure at every tenure, as it must be, since each period gives each claimant present exactly $1/N$; its only defect is the dispersion of realized shortfalls, which is at its largest. (This fixes where the argument with the ex ante theorist has to take place: the forgetting rule’s defect is entirely ex post, and Section 7.4 addresses that theorist directly.) Under the shortfall rule at $\kappa = 1$, expected share stays within two or three per cent of proportional at every tenure, because a newcomer’s shortfall is zero, the shortfalls of the incumbents around her average zero, and surpluses are clipped in only about one claimant-period in thirty; and the dispersion of realized shortfalls falls by two-fifths. This rule satisfies the surrogate constraint, to the accuracy the simulation can resolve. At $\kappa = 2$ the newcomer’s chance is 97 per cent of what her claim warrants, a small but real breach, and dispersion falls by half. At $\kappa = 10$, which is what one gets by pressing the corrective claim at full strength against the very next unit, the newcomer’s chance is little more than half of what her claim warrants.

Table 1: Ten claimants with equal claims, one unit per period, each claimant leaving with probability 0.1 per period and replaced. Chances and expected shares are multiples of the proportional value ($1/N$ per period); $\pm$ gives one standard error across runs (40 runs; 120 for $\kappa = 1$; the first column's standard errors are below 0.005). Cells for tenure 1, 3, 10 and 30 are based on roughly 5,100, 4,200, 2,000 and 230 leavers per 40 runs. Dispersion is the standard deviation, in units, of a claimant's shortfall at the moment she leaves. Details and sensitivity checks in the Appendix.

Rule	Newcomer's					Dispersion of realized shortfall
	first-period chance	Expected share, tenure 1	Tenure 3	Tenure 10	Tenure 30	
Forgetting ($\kappa = 0$)	1.00	0.98 $\pm$.04	1.03 $\pm$.03	1.02 $\pm$.02	0.98 $\pm$.04	0.95
Shortfall, $\kappa = 1$	1.01	0.96 $\pm$.02	0.99 $\pm$.02	1.02 $\pm$.01	1.02 $\pm$.01	0.58
Shortfall, $\kappa = 2$	0.97	0.97 $\pm$.03	1.00 $\pm$.03	0.99 $\pm$.01	1.02 $\pm$.02	0.49
Shortfall, $\kappa = 10$	0.56	0.56 $\pm$.03	0.96 $\pm$.02	0.94 $\pm$.01	0.97 $\pm$.01	0.41
Points for losses, linear (Arizona)	0.26	0.25 $\pm$.02	0.48 $\pm$.02	0.97 $\pm$.02	1.11 $\pm$.03	0.70
Points for losses, squared (Nevada)	0.08	0.08 $\pm$.01	0.23 $\pm$.01	1.01 $\pm$.01	1.22 $\pm$.02	0.58
Queue / preference points (Colorado)	0.00	0.00	0.00	1.01	1.28 $\pm$.01	0.46

The rules in use breach the constraint outright. A linear point system gives a newcomer a quarter of the chance her claim warrants and a claimant who happens to stay thirty periods 11 per cent more than her share; squared points cut the newcomer to a twelfth; a queue gives her nothing, and anyone who leaves within a few periods leaves with nothing. With a larger pool the effect is more severe: with a hundred claimants and a mean tenure of twenty periods, linear points give a newcomer 7 per cent of her proportional chance and a queue gives anyone with less than thirty periods' tenure essentially nothing (Appendix). These rules reduce dispersion, some by more than the moderate shortfall rules, but they do it by transferring expected share from claimants whose tenure turns out short to claimants whose tenure turns out long, which is to deny the short-stayers the surrogate they are owed.¹¹

¹¹Huesch and Brady's subjects, who favoured excluding the first winner, were designing a raffle among a fixed ten with

So, in the recurrence structure, the third claim is established. Memory is required; the forgetting rule is dominated at no cost in anyone’s expected share. But memory may press only as hard as the surrogate constraint allows; to remember lexically is to let an incumbent’s accumulated claim override a newcomer’s present one, and “weaker claims must not simply be overridden by stronger ones.” When the pool turns over, fairness requires a rule that raises a claimant’s chances with her shortfall at a rate that corrects it over roughly the period in which the practice would ordinarily deliver her share; the region around $\kappa = 1$ has this property, and the rules in use lie well outside it. It should be said plainly what this achieves: with the pool turning over, a proportionate rule roughly halves the dispersion of outcomes the forgetting rule produces, and a residual lottery remains. That is not a defeat. The residual is exactly the surrogate the short-stayers are owed, and a rule that eliminated it would take from them what fairness requires they have. Only where the pool is fixed, and everyone will be present for the division, is the residual zero, and there the rule is rotation.

5.4 Bottlenecks and waiting

In the bottleneck structure, where winners leave, nobody accumulates more than one unit, and what varies from claimant to claimant is not units received but time waited. Fungibility fails, because each claimant’s claim is to one unit and to having it soon. What is divisible is the waiting.

Under the forgetting rule, waiting is itself raffled. With a pool of ten kept full by arrivals and one unit per period, some are served at once and about one in seventy waits forty periods or more, though everyone’s expected wait is nine. A queue serves everyone at nine. Waiting is a bad, claims to avoid it are equal, and time, unlike a flat, is divisible. So the Division Principle applies to the waiting, and it says: equalize it. This is why John and Millum (2020) are right that a queue is often fairer than a lottery, and why Hersch and Rowe (2024) are right to distinguish the “bottleneck” cases where queues are appropriate from the “scarcity” cases where lotteries are: in a bottleneck everyone will be served and what is divisible is the delay; in a one-shot scarcity nothing is divisible; between them lies the recurrence structure, where what is divisible is the units. It is also why the queue is the right kind of rule in the bottleneck structure though the wrong kind in the recurrence structure. There a claimant’s claim is to a share of the units, accruing period by period, and a queue gives a short-stayer none of hers; here her claim is to one unit and to not being kept waiting longer than others, and, where everyone will be served, a queue gives everyone exactly that. The newcomer at the back is not being overridden; her claim to be served *now* would be a claim to jump the queue.

The surrogate constraint carries over, and it qualifies the queue as it qualified the shortfall rule. Where everyone will be served, the division of the waiting is available to all, and the queue completes it. But where claimants may leave unserved, by dying, losing eligibility, or giving up, and the allocator cannot tell who, those to whom the division is not available are owed the surrogate: a chance, in each period, proportional to their claim. A pure queue denies it: everyone who leaves unserved leaves having been given no chance at all. So the fair rule in a bottleneck with unforeseeable attrition is a waiting-

no newcomers, where exclusion is what the Division Principle recommends; their verdict is not evidence for exclusion where the pool turns over.

weighted lottery, in which chances rise with time waited at a rate that keeps every claimant's expected treatment proportional over whatever stay she turns out to have. This is the bottleneck counterpart of the moderate shortfall rule, and several real systems are of this kind: the Grand Canyon's tickets-for-years and Arizona's extra numbers are waiting-weighted lotteries, not queues. Only where exit is negligible is the pure queue the fair rule. And where an exit is foreseeable and imminent, the claim is urgent, and whatever rule is used must be weighted for it, as transplant allocation is; where claims differ in strength, waiting should likewise be weighted by claim.¹²

The verdict for the visa lotteries follows. They are bottlenecks with attrition: a selected worker leaves the pool, and an unselected one may lose her sponsor or her status. The fair rule is a lottery weighted by years of unsuccessful registration, and by urgency and the strength of claims. It is not the forgetting rule, and it is not the rule that took effect in 2026, which weights registrations by the wage level of the job, a reason of usefulness rather than a claim (Section 2): that reform made the lottery more responsive to the employer's interest and left it blind to the applicant's history. The quantity to weight by is on the record the government already keeps: the number of years each unique beneficiary has registered and been passed over. Hersch and Rowe (2024, secs. 1, 3) reach the opposite verdict: since demand for visas will never be met, they classify the case as scarcity rather than bottleneck and endorse the lottery. But their categories are built for a single allocation, and they say themselves that in a bottleneck a periodic lottery is unfair because a person "who arrives into the lottery pool early and is continuously unlucky" may wait far longer than a lucky newcomer (sec. 4). That is true whether or not everyone will eventually be served; what makes it true is that the same claimants meet the same allocator year after year, and the visa lotteries have that feature whichever category they fall in.

6 How far does memory reach?

Any view on which past allocations bear on present claims must say how far the bearing extends. If Nevada must remember Maggie's 2026 loss in 2027, must it remember her loss in the housing lottery? Her loss in Arizona's draw? Her mother's? A view that cannot stop somewhere principled collapses into a general demand that luck be equalized, which is a different and far more contested thesis. In this section I derive the bounds on memory from the argument that generates it. There are four.

6.1 Memory is bound to the practice

The corrective claim of Section 4 arises from a specific fact: that a particular agency's allocation of a particular kind of good left residual unfairness because its units could not be divided. The claim is a claim on the division of *those* units, addressed to *that* agency, and both restrictions do work.

The good matters because the 2027 tag can be used to divide the sequence of tags only because it is a tag. It cannot be used to divide the sequence of flats, because a flat is not a portion of the good the

¹²The economics of dynamic allocation reaches related conclusions by a different route: Leshno (2022) shows that in an overloaded waiting list a simple randomized assignment policy can reduce the misallocation that strict waiting-time priority produces, and Arnosti and Shi (2020) compare lotteries and wait-lists for affordable housing. Those results concern efficiency; the argument here concerns what the applicants are owed.

hunter has a claim to. The Division Principle requires that a divisible good be divided; it does not require that shortfalls in one good be made up in another. So a loss in the housing lottery generates no claim in the tag lottery. Goods are individuated by claims, not by names: a Nevada sheep tag and an Arizona sheep tag are different goods, because a claim to hunt in one state is not satisfied by a tag for the other, whereas two lotteries run by one government for the same visa under different labels would be one good, and the shortfall would carry across them. An allocator cannot shed its obligation by splitting a practice in two, because the claims that fix what the good is do not split with it.

The agency matters because the corrective claim is a claim that a division be completed, and a division is something an agent does with what it has. Where two independent allocators divide units of one kind among overlapping claimants, each is bound by the division it is making and neither by the other's: a claimant's shortfall with the first is a fact about how the first has treated her, and the second has none of the first's units to give. The case is narrow, since claims rarely range indifferently over two allocators' units; but where it arises the bound holds, because the Division Principle constrains the divider.

This is what distinguishes the thesis from luck egalitarianism, which holds that it is bad, or unjust, for one person to be worse off than another through no fault of her own (Dworkin 1981; Cohen 1989; Temkin 1993). That thesis concerns welfare or resources in general, and its currency is compensation: a shortfall in one good can be made up in another. Mine concerns the fair division of a specific good by the agency dividing it, in the good's own currency, and says nothing about whether the persistent loser is worse off overall. Nor is she a gambler whose losses are her own affair, in Dworkin's sense: she did not choose to gamble; the lottery was imposed as a second best to the division fairness required, and its results are the residue of that second best, not option luck.

6.2 Memory is bound to the claim

The corrective claim is a strengthened claim to the good, not a credit the claimant holds. It lasts exactly as long as her claim to the good lasts. Someone who applied unsuccessfully for a tag for ten years and then gave up hunting is owed nothing; there is no tag she has a claim to, so there is nothing her shortfall could raise her chances at. Memory is not compensation. Compensation is what one owes when a wrong has been done and its effects must be undone; what the lottery owes its persistent losers is a fair division of the good it is still dividing.

Four questions about the claim's boundaries have determinate answers on this view. *Transition*: when a forgetting practice adopts memory, do losses under the old rule count? Yes. A claimant's shortfall existed whether or not the rule tracked it, and a practice that begins to track it has no principled reason to start the count at zero; this is the opposite of what a rectificatory view would say, since under the old rule no one did anything wrong. *Absence*: does a period in which a claimant did not apply count toward her share? No. She asserted no claim on that period's unit, so her share to date is the sum over the periods in which she did; Nevada's rule that points lapse after two years of not applying is, in this respect, right. *Time-varying claims*: what is a claimant's shortfall when her claim was weak in 2026 and strong in 2028? Her share to date is the sum, over the periods in which she had a claim,

of her claim in that period as a fraction of all claims present, and her shortfall is that sum less what she received; this is how weighting by the strength of claims and memory combine, since both are inputs to one quantity. *Regress*: if the 2027 allocation must correct the 2026 shortfall, must the 2028 allocation correct the 2027 allocation's failure to correct it fully? No. Shortfall is a state, not a stack of debts: at each period there is one quantity, the difference between what the claimant has received and her share to date, and correcting it is the whole of what is owed.

6.3 Memory is bound to the claimant

Whose history? The claimant's, and the claimant is by default the person, because claims are duties owed to someone and a person is who they are owed to. This is not a matter the practice may settle as it likes: New Mexico cannot make its persistent losers vanish by declaring that its claimants are "this year's applicants," a new set each year, since the person who applied last year and this is one person with one claim, and it is she to whom the undertaking was made. What a practice may do is pool the claims of several persons into one claimant, as housing systems do when they register households and as hereditary allotments once did with lineages; then the pooled claimant's history is the pool's, and persists as the pool does, through changes in its membership. I accept the implication for lineages. It follows from a decision about whose claims are pooled, not from any licence to define persisting claimants out of existence.

What the thesis does not say is that there is a general obligation to correct the shortfalls of one's ancestors. Claims of that kind, where they exist, are claims of rectification, and they arise from wrongs: takings, exclusions, breaches. The argument of this paper begins from the Division Principle, which governs the fair division of a good one is dividing, and goes no further than that principle goes. A practice that divided its good fairly, by rotation, owes nothing to anyone's descendants; a practice that raffled it owes its present claimants a division of what remains; and a practice that stole it owes something else, on grounds this paper does not supply.

6.4 Memory is bound in weight

The fourth bound is the one Section 5 established, and it can be stated in a sentence. Memory may press a claimant's shortfall only as hard as the surrogate constraint allows: never so hard that anyone who may leave before the division is complete is denied the chance her present claim warrants. Where everyone will stay, that is no bound at all, and the rule is rotation or the queue; where the pool turns over, it is the bound that separates the moderate rules from the lexical ones.

7 Objections

7.1 Losing on purpose

Every allocator that has considered memory has met the objection that it rewards losing, and the objection is serious. The NBA's draft lottery gives the worst teams the best chance at the best players, and the predictable result was that teams lost on purpose; the league flattened the odds in 2019, so that

the three worst records now share a 14 per cent chance at the first pick rather than the worst holding 25 per cent, for exactly that reason.¹³ If Nevada's squared points are generous enough, someone might enter the sheep draw for years with no intention of hunting sheep, to build points. If the visa lottery counted years of registration, employers might register workers they had no intention of hiring.

Two things should be said, and one conceded. First, the objection presupposes that losing is under the claimant's control, and in the cases that motivate this paper it is not: an applicant cannot make herself lose; she can only apply. What she can do is apply without a genuine claim, and that is the problem of verifying claims, which every allocation faces. Second, where losing *is* under the claimant's control, as in a sports league, the objection shows that fairness is in tension with incentive, which is a reason to weigh fairness against other things, not a reason to think fairness itself requires forgetting; the NBA did not conclude that a memoryless draft would be fairer, but that some fairness was worth trading for competitive integrity. What must be conceded is that memory raises the value of a manufactured claim: under the forgetting rule a speculative entrant gains one ticket, under a rule with memory a ticket that grows. A practice that cannot verify claims therefore has more to lose from memory than one that can, and the right response is to attach memory to participation that costs something, as Nevada does by requiring a licence to be bought each year, and as the visa lottery could do through the registration fee it already charges each employer for each beneficiary, \$215 since 2025. This is a constraint on implementing fairness, sometimes a serious one; it is not evidence that fairness has no memory.

7.2 Expression

Wasserman (1996), Stone (2011), Burgers (2016), and Spiekermann (2022) hold, in different ways, that what makes a lottery fair is not what it distributes but what it expresses or excludes: equal standing, impartiality, the absence of any reason that could justify preferring one claimant to another. On such a view a rule with memory marks people. The reply is that past shortfall is not a reason of the kind such views exclude. Stone's lotteries are just because they keep illegitimate reasons out of a decision no legitimate reason can settle; but a claimant's shortfall from her share of the very good being allocated is the paradigm of a legitimate, comparative reason, and a procedure that excludes it excludes a good reason along with the bad. What the forgetting rule expresses to a claimant who has entered it four times is not equal standing but that the agency has kept no record of her. And Spiekermann, in setting out a case, stipulates that a procedure whose outcome is fixed across occasions must not be reused, since "a repeated use of the procedure is unfair" (Spiekermann 2022, 110 n. 2); that is a narrower point than mine, but it grants that what happened at the first occasion can bear on the fairness of the procedure at the second.

¹³NBA, "NBA Draft Lottery: How it works" (2026). Elster (1988) noted the incentive problem with the earlier version of the draft.

7.3 Equality, not fairness

Someone might grant everything I have said about the pattern of outcomes and deny that fairness is what condemns it. What is bad about the forgetting rule, on this suggestion, is that it produces inequality, or leaves some badly off, and those are matters for a theory of equality or of priority to the worse off (Parfit 2000), not of fairness. Hold welfare fixed, and the difference appears. Suppose the persistent loser is, all things considered, no worse off than the repeat winner, because she has other goods the winner lacks. Her complaint survives. It is not that she is badly off but that this agency, dividing this good among people with equal claims, has given her less than her share and others more. The complaint is comparative, it concerns the good in question, and it is addressed to the allocator: the marks of a fairness complaint in Broome's sense. A priority view would often recommend memory too, but for reasons that would equally recommend giving the loser something else, and it could not explain why the claim dies with her interest in the good or why it is owed by the allocator rather than by anyone in a position to help.

7.4 Ex ante

The last objection comes from the view, originating in Diamond's (1967) reply to Harsanyi (1955) and defended most carefully by Frick (2015; see also Otsuka and Voorhoeve 2009), that what matters to the fairness of a procedure is the prospects it gives people at the time of choice, not the outcomes it produces. On that view the forgetting rule and rotation are equally fair, because at the outset each claimant has the same prospects under both, and the ex post dispersion I have made so much of is how equal prospects happened to resolve. Section 5 conceded the premise: in a fixed population the forgetting rule's defect is entirely ex post.

Two replies. The first is that even the ex ante view must say when the ex ante is. If it is the beginning of the practice, the view cannot prefer the forgetting rule to rotation, since they are ex ante identical, and the Division Principle settles the question it leaves open. If it is the beginning of each period, so that fairness requires equal chances at each unit *whatever has gone before*, the view is asserting separability with respect to history-blind claims, which is what the cake refutes. There is no third setting of the clock at which the ex ante view both distinguishes the two rules and favours the one that forgets.

The second reply is that the ex ante view was developed for one-shot cases, where an outcome cannot be undone and prospects are all a procedure can distribute: a vaccine given once, a rescue made once. In the recurrence structure the outcome of any one period is not final, because the sequence is still being divided, and what was a prospect in one period becomes a fact the next period's allocation can attend to. Nothing in the motivation for ex ante evaluation carries over to a structure in which the allocator will have further units to give and knows who has received what.

8 Conclusion

Fairness has memory. What the memory tracks is a claimant's shortfall from her fair share of the good being divided, in units where later units are as good as earlier ones and in waiting where they are not;

it is owed by the agency dividing the good, for as long as she has a claim to it; and it may be pressed only as hard as the surrogate every claimant is owed allows. Each clause was derived from a single principle the standard theory already contains, that a divisible good should be divided, together with the observation that a good allocated repeatedly is divisible whether or not its units are.

New Mexico’s draw and the Wimbledon ballot are unfair as designed, not because they are lotteries but because they forget; so are the visa lotteries, and the 2026 reform made them responsive to usefulness rather than to claims. Preference-point queues of the Colorado and Wyoming kind are the right rule in the wrong structure. A shortfall rule is not exotic: it is what a household does when it keeps a rota, and what a divisor method does for parliaments, extended to units that arrive one at a time.

The consequence for theory is that fairness, on the best account we have of it, is not separable across occasions unless claims are allowed to carry history, and it should have been visible from the start in the example Broome chose to illustrate the Division Principle: military service, divided by conscripting everyone for a short time. Everyone’s short time is a rotation. The lottery is what one does when one cannot rotate, and one can rotate more often than the lottery’s defenders have noticed.

Appendix: simulations

All figures reported in Section 3 and Section 5 were obtained as follows. There are N claimants with equal claims and one indivisible unit per period. At each period, claimant i receives the unit with probability proportional to a weight ω_i . The rules are: forgetting, $\omega_i = 1$; shortfall with parameter κ , $\omega_i = \max(0, 1 + \kappa(r_i/N - w_i))$, where r_i is the number of periods i has been present and w_i the number of units she has received; linear points, $\omega_i = 1 + \ell_i$, where ℓ_i is the number of periods since i last received a unit; squared points, $\omega_i = 1 + \ell_i^2$; queue, $\omega_i = 1$ for the claimants with the largest ℓ_i and 0 otherwise; rotation, a uniformly random order drawn once and then cycled.

In the fixed-population runs, the same $N = 10$ claimants are present for $T = 10$ or $T = 40$ periods; each rule was run 6,000 times for $T = 10$ and 3,000 times for $T = 40$. The analytic values for the forgetting rule agree with the simulated values to two decimal places.

In the churn runs (Table 1), each claimant leaves at the end of each period with probability q , independently of her history, and is replaced by a newcomer with $r_i = w_i = \ell_i = 0$; the horizon is 1,500 periods with the first 200 discarded, and each rule was run 40 times at $N = 10$, $q = 0.1$ (120 times for $\kappa = 1$). For each claimant who leaves, her tenure, total, and shortfall are recorded. “Expected share, tenure L ” is the mean total among claimants whose tenure was exactly L , divided by L/N ; standard errors are across runs. “Newcomer’s first-period chance” is the mean, over newcomers, of their probability of receiving the unit in the period they arrive, times N . For the shortfall rule this probability is $N/(1 + \sum_j \max(0, 1 + \kappa d_j))$, summing over the incumbents’ shortfalls d_j ; without the floor the sum would be $(N - 1) + \kappa \sum_j d_j$, whose expectation is $N - 1$, giving the newcomer $1/N$ at every κ . The floor raises the sum by clipping surpluses, and the clipping grows with κ : at $\kappa = 1$ about 3.6 per cent of claimant-periods are clipped, at $\kappa = 2$ about 14 per cent.

Sensitivity. At $N = 10$ with $q = 0.05$ (mean tenure 20) the newcomer's chance, as a multiple of $1/N$, is 1.01 at $\kappa = 1$, 0.98 at $\kappa = 2$, 0.55 at $\kappa = 10$, 0.21 for linear points, 0.05 for squared points and 0 for the queue; at $q = 0.2$ (mean tenure 5) the corresponding figures are 1.01, 0.97, 0.57, 0.36, 0.16 and 0. At $N = 100$ with $q = 0.05$ (six runs of 3,000 periods) they are 1.00 at $\kappa = 1$, 0.94 at $\kappa = 2$, 0.07 for linear points and 0 for the queue, and under the queue expected share is 0.00 at tenures 1, 3 and 10 and 0.06 at tenure 30. The dispersion of realized shortfall falls under every rule with memory in every configuration. The simulation code (`simulations.py`) accompanies this paper.

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